LAB 4 — THINKING SHEET

Regularization and Feature Selection

This sheet is for you. Preparing for the exam starts here.

I recommend working in pairs, trying to explain your point of view to each other.

Running the code is useful, but it is not the goal of this lab. The goal is to practise predicting, observing, explaining, and experimenting.

For each investigation: PREDICT → RUN & OBSERVE → EXPLAIN → EXPERIMENT

Write short answers. A few words, a sketch, an equation, or a small graph may be enough.

1 — Least Squares baseline

PREDICT: With only 60 training examples, what difficulties might an unregularized linear model face?

With only 60 samples, an unregularized linear model just minimizes L_2, without considering the complexity given by $\|w\|$. I might overfit

OBSERVE / EXPLAIN: Compare training and test performance. What does the gap suggest?

The test error is 10x the training error → it generalizes poorly (overfits)

2 — Lasso and sparsity

PREDICT: As Lasso regularization α increases, what happens to coefficient magnitudes and the number of nonzero coefficients?

The coefficients magnitud decreases as well as the # of nonzero coefficients. The validation error decreases until the "high importance" of complexity makes the model underfit

OBSERVE / EXPLAIN: What does the Lasso path show? Does a sparser model necessarily predict better?

The lasso path shows us how to achieve the desired balance between low complexity and low L_2. No, it may underfit!

EXPERIMENT: Change the α grid. What did you expect, and what did you observe?

High α-min → sparse $\vec{w}$, high L_2, low α-max → non-sparse $\vec{w}$, low L_2,

3 — Cross-validation: selecting α

PREDICT: What might happen to validation error for very small or very large α?

for both: LV increases

very small α → you're considering just L_2 → if complexity is high you might overfit. very large α → L_2 → you're not considering you might underfit

OBSERVE / EXPLAIN: Which α minimizes mean validation MSE? Why might it change with a different random training sample?

$\alpha = 13219$. (?)

Why must scaling be fitted inside each CV training fold, and why must the final test set remain untouched?

Because each fold represent a different trained model so it has it's own scaling. Because you can't train over it or you won't get an accurate estimation of L_2

4 — Ridge vs Lasso

PREDICT: Which method can set coefficients exactly to zero? How does Ridge typically affect them?

OBSERVE / EXPLAIN: Compare Least Squares, Lasso and Ridge coefficients and validation results. What surprised you?

EXPERIMENT: Choose one coefficient or α value. What changed, and why?

5 — Final test evaluation

PREDICT: Which model will perform best on the held-out test set? Give one reason.

OBSERVE / EXPLAIN: Compare test R^2 and MSE. What do they tell you? Is the CV winner necessarily the test winner?

Remember: higher R^2 and lower MSE are better when comparing models on the same test set.